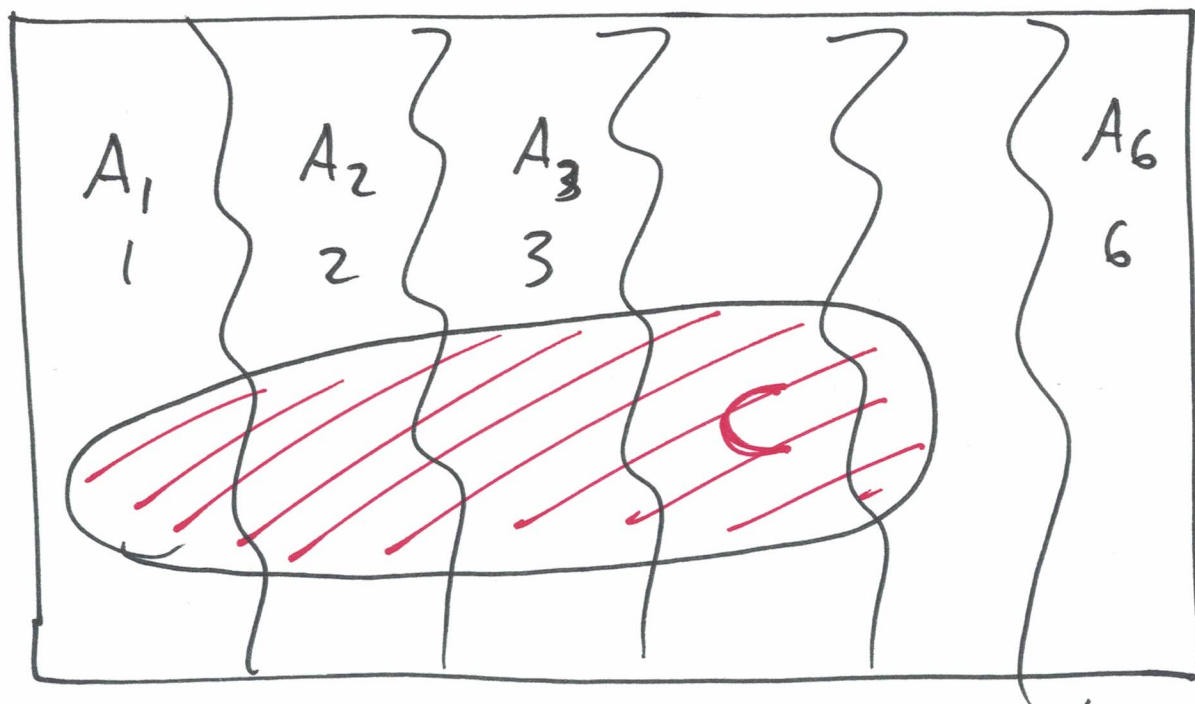
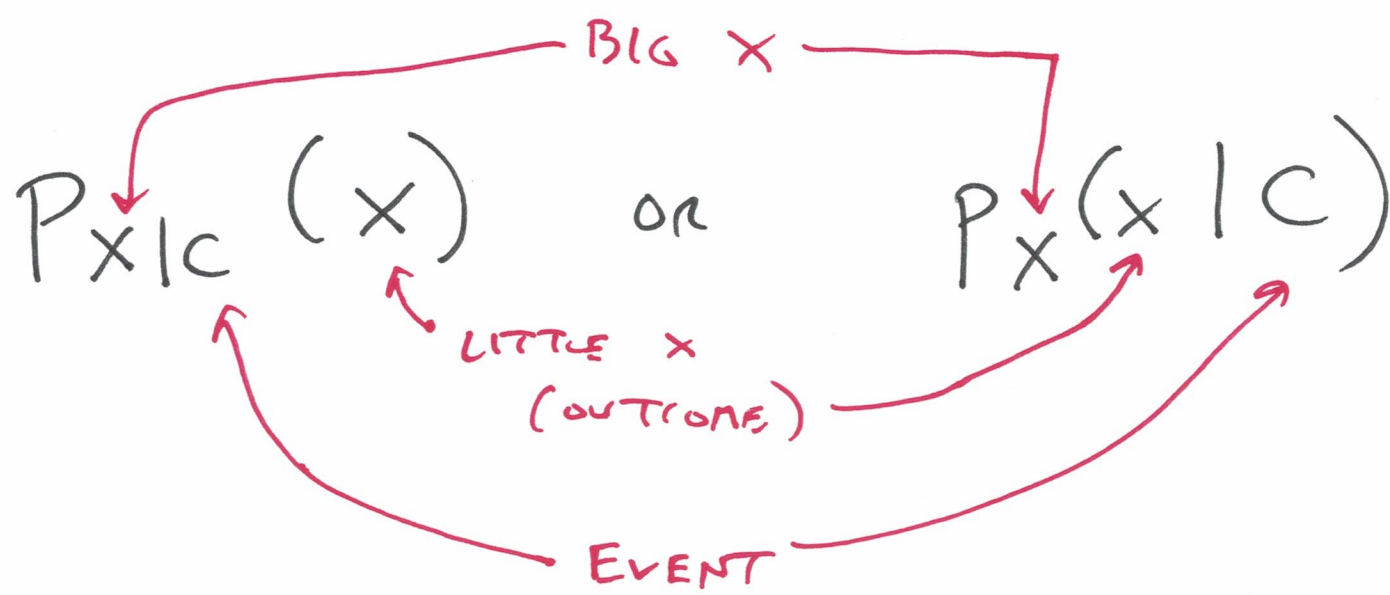


CONDITIONAL PMFs



→ $P_{X|C}(x) ?$

CONDITIONED
ON
EVENT C



EX X BINOMIAL WITH $n=6$ AND $p = \frac{1}{3}$.

WHAT IS $P_X(x | X > 2)$?

ORIGINAL PMF:

$$P_X(k) = \binom{6}{k} \left(\frac{1}{3}\right)^k \left(\frac{2}{3}\right)^{6-k}$$

$k = 0, 1, \dots, 6$

$$P_X(0) = \left(\frac{2}{3}\right)^6 = \frac{64}{729}$$

$$P_X(1) = 6\left(\frac{1}{3}\right)\left(\frac{2}{3}\right)^5 = \frac{192}{729}$$

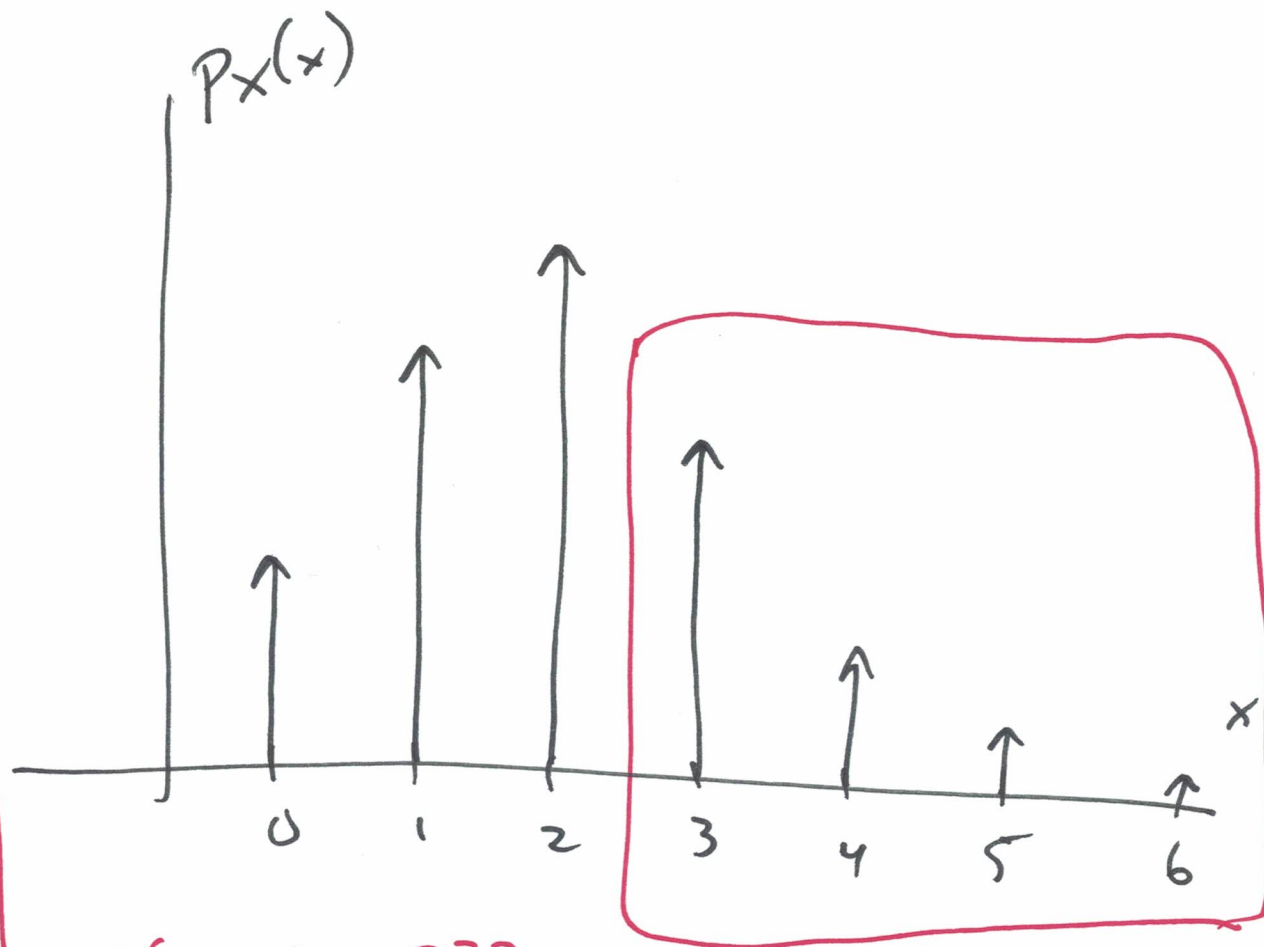
$$P_X(2) = \frac{240}{729}$$

$$P_X(3) = \frac{160}{729}$$

$$P_X(4) = \frac{60}{729}$$

$$P_X(5) = \frac{12}{729}$$

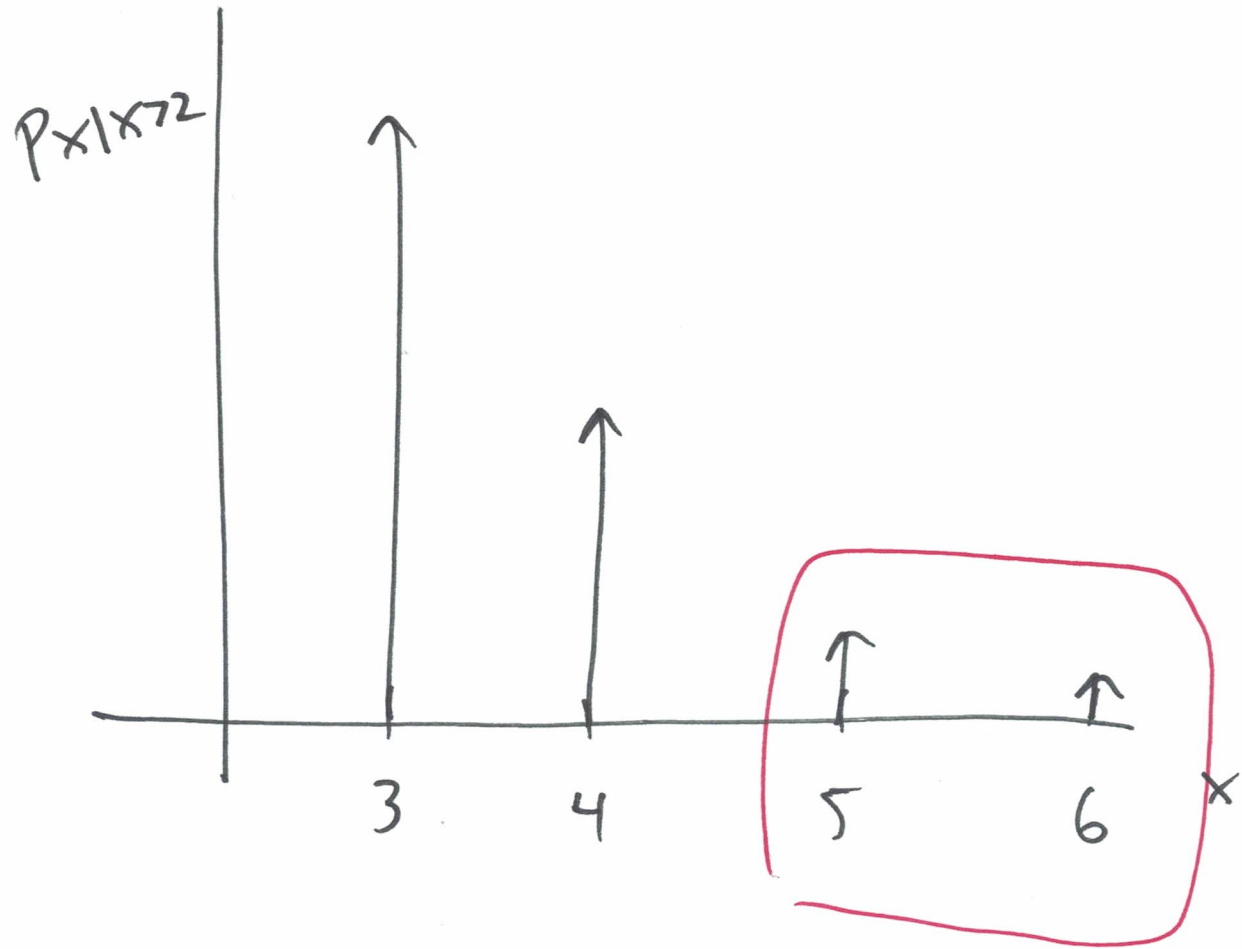
$$P_X(6) = \frac{1}{729}$$



$$P(X > 2) = \frac{233}{729}$$

$$P_{X|X>2}(k) = \frac{P(X=k \text{ AND } X>2)}{P(X>2)}$$

k	$P_{X X>2}(k)$
3	$\frac{160}{233}$
4	$\frac{60}{233}$
5	$\frac{12}{233}$
6	$\frac{1}{233}$



$$P(X > 4 \mid X > 2) = \frac{13}{233}$$

$$E(X \mid X > 2)$$

$$= 3 \cdot \frac{160}{233}$$

$$+ 4 \cdot \frac{60}{233}$$

$$+ 5 \cdot \frac{12}{233}$$

$$+ 6 \cdot \frac{1}{233}$$

$$= \frac{786}{233} = 3.373$$

$$\text{Var}(X \mid X > 2)$$

$$= 3^2 \cdot \frac{160}{233} + \dots + 6^2 \cdot \frac{1}{233} - (3.373)^2$$

USEFUL PROPERTY:

$$E(X) = \sum_{i=1}^k E(X|B_i) P(B_i)$$

 $\{B_i\}$ PARTITION
THE SAMPLE
SPACE

EX $H = \text{HIGH BLOOD PRESSURE}$
 $N = \text{NORMAL}$

$$E(X) = E(X|H) P(H) + E(X|N) P(N).$$

SPECIAL CASE:

$$E(NX)$$

"
Y

WHERE N INTEGER,
 N AND X ARE INDEPENDENT.

$$E(Y|N=i) P(N=i)$$

" $P_N(i)$

e.g.

$N = \#$
STUDENTS

$X = \text{SERVICE}$
TIME

$$E(Y) = \sum_{i=0}^{\infty} \underbrace{E(Y|N)}_{E(NX|N)} P(N)$$

$$= \sum_{i=0}^{\infty} N \cdot E(X) P(N)$$

$$= E(X) \underbrace{\sum_{i=0}^{\infty} N P(N)}$$

$$= E(X) E(N)$$

$$E(Y) = \sum_{n=0}^{\infty} E(Y|N=n) P(N=n)$$

$p_N(n)$

$$= \sum_{n=0}^{\infty} E(NX|N=n) p_N(n)$$

$$= \sum_{n=0}^{\infty} E(nX) p_N(n)$$

$$= \sum_{n=0}^{\infty} n E(X) p_N(n)$$

$$= E(X) \sum_{n=0}^{\infty} n p_N(n)$$

$$= E(X) E(N)$$

OH SO DOES THAT MEAN

$$\text{Var}(X) = \text{Var}(X|H)P(H) + \text{Var}(X|N)P(N) ?$$

No! ☹

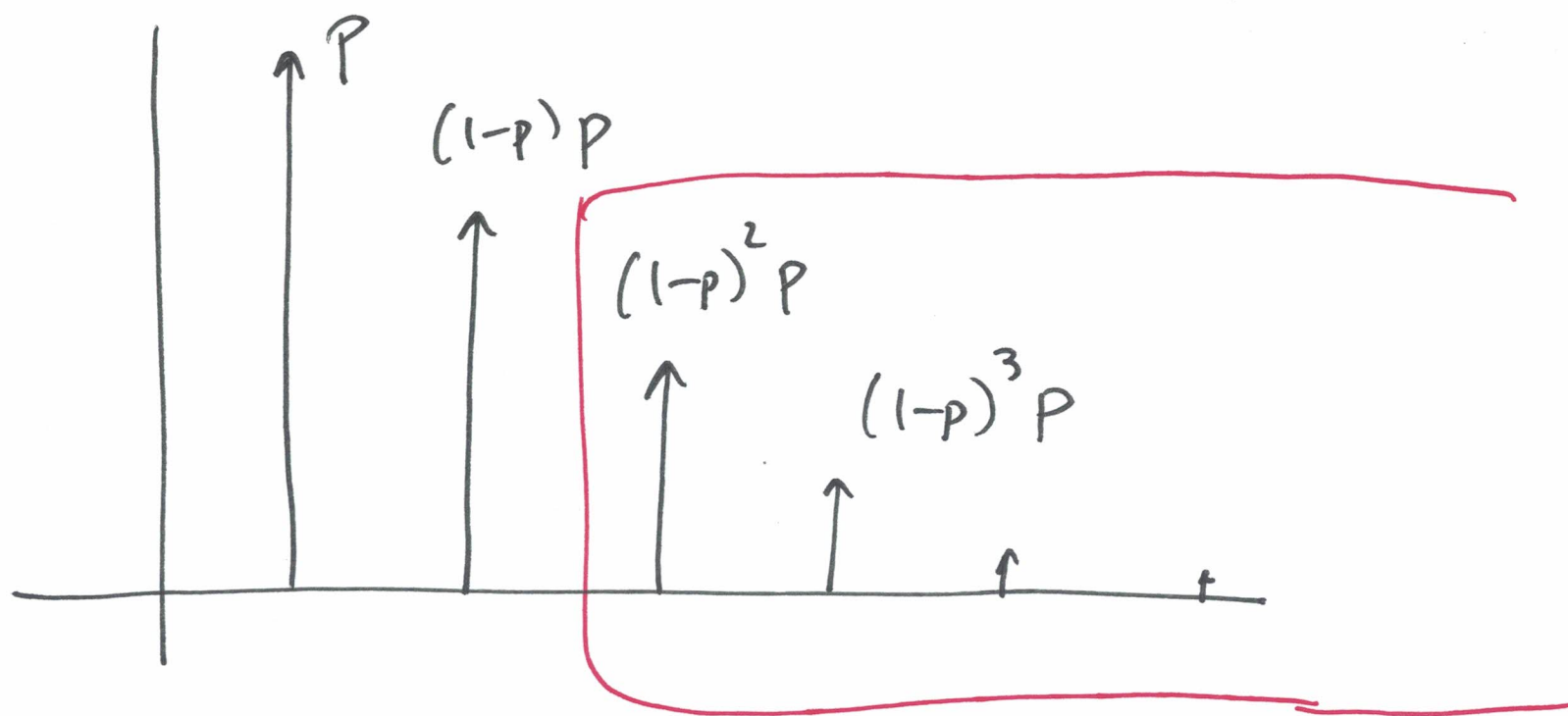
WE CAN ALWAYS SAY

$$E(g(x)) = \sum E(g(x)|B_i) P(B_i)$$

BUT ~~$g(x)$~~ IS NOT IN THE FORM $g(x)$
 $\text{Var}(X)$ $\text{Var}(X|H)$ AND $\text{Var}(X|N)$
DEPEND ON DIFFERENT MEANS

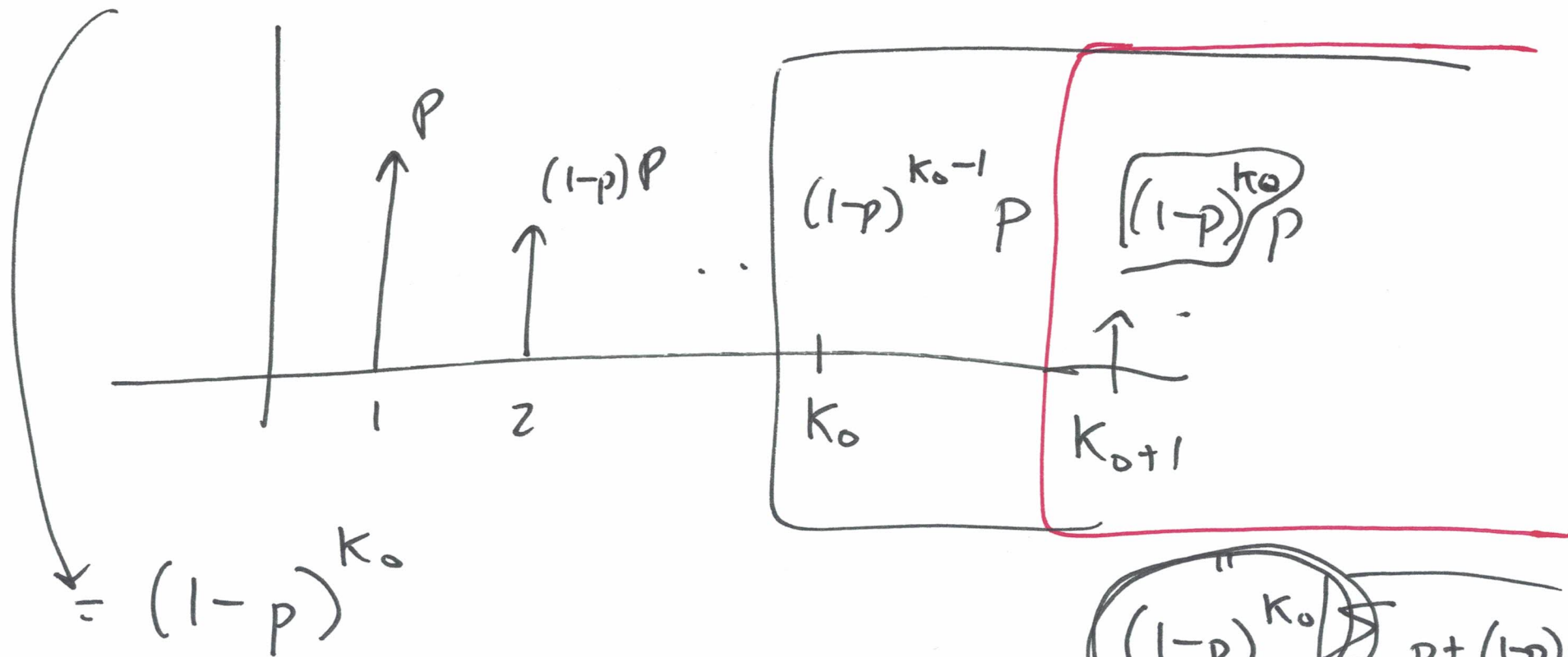
SPECIAL PROPERTY OF GEOMETRIC
RANDOM VARIABLE : MEMORYLESS.

(ONLY TRUE FOR GEOMETRIC)



RED BOX LOOKS JUST LIKE WHOLE THING
AFTER RE-NORMALIZATION.

$$P(X > k_0)$$



$$P(X > k | X > k_0)$$

$$= \frac{(1-p)^{k+k_0}}{(1-p)^{k_0}} = (1-p)^k$$

Sum of red part.

$$(1-p)^{k_0} \left[p + (1-p)p + (1-p)^2 p + \dots \right] = 1$$

$$P(X > k + k_0 \mid X > k_0)$$
$$= P(X > k)$$

$$P(X > 5 \mid X > 2)$$
$$= P(X > 3)$$

POISSON DISTRIBUTION

$$P(X=k) = \frac{\alpha^k}{k!} e^{-\alpha}$$

DOES THIS REALLY SUM TO 1?

$$\sum_{k=0}^{\infty} \frac{\alpha^k}{k!} e^{-\alpha} = e^{-\alpha} \sum_{k=0}^{\infty} \frac{\alpha^k}{k!}$$

$$= e^{-\alpha} \cdot e^{\alpha} \\ = 1.$$

$$1 + \alpha + \frac{1}{2!} \alpha^2 + \frac{1}{3!} \alpha^3 + \dots$$

TAYLOR SERIES

FOR e^{α}

MONTY HALL PROBLEM: 3 DOORS

YOU CHOOSE DOOR 1

HOST OPENS DOOR 3. GOAT.

SHOULD YOU SWITCH TO DOOR 2?

TRUE PRIZE	HOST OPENS	PROB (HOST OPENS TRUE PRIZE)
1	1	$\emptyset$
$\frac{1}{3}$ 1	2	$\frac{1}{2}$ L
1	3	$\frac{1}{2}$ L
2	1	$\emptyset$
$\frac{1}{3}$ 2	2	$\emptyset$
2	3	1 W
3	1	$\emptyset$
$\frac{1}{3}$ 3	2	1 W
3	3	$\emptyset$

$P(\text{WIN BY SWITCHING})$

$$= \frac{\frac{1}{3} + \frac{1}{3}}{\frac{1}{3} + \frac{1}{3} + \frac{1}{3}} = \frac{2}{3}$$